Yiwei Shu

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EDUCATION

University of Chinese Academy of Sciences, Beijing, China 2024 – present
Ph.D. in Robotics and Space Robotics

- Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences.
- Research focus on autonomous exploration, legged field robots, active SLAM, motion planning, and planetary robotics.

ShanghaiTech University, Shanghai, China Sept 2020 – June 2024
B.Eng. in Electronic Information Engineering

University of Illinois Urbana-Champaign, Urbana, IL, USA Dec 2022 – May 2023
Exchange Program

- Quantum Physics, Data Science Discovery, and Statistical Analysis, all completed with A+ grades.

SKILLS

Programming: Python, C, C++, MATLAB, R, Verilog, LaTeX
Robotics: ROS/ROS 2, PCL, Unitree Go2, RM75, STM32, LiDAR, depth sensing
Simulation and Learning: Isaac Lab, MuJoCo, PyTorch, scikit-learn, numerical simulation
Engineering: Linux, Git, SolidWorks, Altium Designer, ANSYS Fluent, Cadence, Vivado

RESEARCH EXPERIENCE

Time-Efficient Multi-Gait Lunar Navigation for Unitree Go2 2026 – present
Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences

- Develop goal-directed SLAM and adaptive locomotion for fast quadruped navigation across GPS-denied lunar terrain.
- Integrate Livox Mid-360 LiDAR-inertial odometry, occupancy-grid mapping, and point-goal planning on the Unitree Go2.
- Build 0.16 g Isaac Lab benchmarks and gait-conditioned models for time-to-goal, traversal risk, and sim-to-real evaluation.

Reactive Whole-Arm Manipulation and Pose-Constrained Grasp Planning 2025 – present
Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences

- Recover configuration-space waypoints from end-effector targets using whole-arm clearance, orientation, and joint-feasibility constraints.
- Develop reachability- and manipulability-aware two-stage grasp planning for objects near robot workspace boundaries.
- Validate on RM75, Franka Panda, UR5e, and KUKA iiwa platforms in simulation and real-robot experiments.

Autonomous Fixed-Wing Target Landing 2023 – 2026
ShanghaiTech Automation and Robotics Center (STAR Center)

- Designed and self-built a low-cost underactuated fixed-wing platform, including avionics, sensing, and launch hardware.
- Combined CFD-based aerodynamic characterization, IMU guidance, monocular visual servoing, and finite-state control.
- Validated the full system through randomized simulation and physical flight experiments.

SLAM-Point Cloud Matching Sept 2022 – Dec 2022
ShanghaiTech Automation and Robotics Center (STAR Center)

- Reconstructed registered 3D maps from raw LiDAR point clouds.
- Implemented ICP-based pose estimation and point-cloud alignment using the Point Cloud Library (PCL).

Learning-Based Medical Imaging Reconstruction Sept 2021 – July 2022
Smart Medical Information Research Center, ShanghaiTech University

- Designed U-Net feature extraction and fusion for photoacoustic image denoising, achieving test MSE below 10e-4.
- Developed and tuned a TAT-Net pipeline for sinogram-to-image reconstruction from thermoacoustic measurements.

MANUSCRIPTS UNDER REVIEW

Finite-State Visual Servo Control for Autonomous Target Landing of Low-Cost Underactuated Fixed-Wing Aircraft 2026
Yiwei Shu, Shuquan Wang. *Manuscript under review at Aerospace Science and Technology*.

From End-Effector Targets to Whole-Arm Postures: Configuration-Space Waypoint Recovery for Reactive Manipulation 2026
Yiwei Shu, et al. *Manuscript under review at IEEE Robotics and Automation Letters*.

RKILPlanner: Risk-Based Kinematic Imitation Learning Planning Towards Human-Like Driving in Highway Scenarios 2026
Yiwei Shu, Shuquan Wang. *Manuscript submitted to ROBIO 2026*.

PUBLICATIONS

Expanding Grasp Reachability: Field-Based Pose-Constrained Two-Stage Grasp Planning 2026
Yiwei Shu, Shuquan Wang, Jun Li, Hanying Sang, Yidong Ye. *5th International Conference on Service Robotics (ICoSR 2026), accepted full paper and oral presentation*.

Learning-Based Extended-Workspace Manipulation: A Self-Relocating Space Crawler for Autonomous Deep-Space Outposts 2026
Yiwei Shu, Shuquan Wang. *46th COSPAR Scientific Assembly, poster presentation*.

Attitude Control of Unpowered Aircraft in Flight Based on Kalman Filter Visual Servo-PID Optimization Algorithm 2025
Yiwei Shu, Shuquan Wang. *Proceedings of the 44th Chinese Control Conference (CCC 2025)*.

LEADERSHIP AND SERVICE

UCAS Student Union 2024 – present
Department Head, Student Life and Culture

- Lead department operations and coordinate campus cultural and student-life initiatives.

ShanghaiTech RoboMaster Team 2021 – 2023
Sentry Team Lead

- Coordinated embedded programming, PCB design, mechanical modeling, and team planning for the autonomous sentry robot.

ShanghaiTech University Sept 2022 – Dec 2022
Teaching Assistant, Digital Circuits

- Led tutorials, answered technical questions, and assisted with assignment and examination grading.

ShanghaiTech University Social Practice Program July 2021
Volunteer Teacher

- Taught at a local primary school and conducted field research on local education conditions.

HONORS AND AWARDS

- Outstanding Student Award, University of Chinese Academy of Sciences (2025-2026)
- Excellence Rising Star Award, UCAS Student Union (2025-2026; 2024-2025)
- Outstanding Student Leader Award, University of Chinese Academy of Sciences (2024-2025)
- Third Prize, RoboMaster National Competition (2022)
- Second Prize, East China Division, 6th National Undergraduate IC Innovation and Entrepreneurship Competition (2022)